Clinton Power Station
8401 Power Road
Clinton, IL 61727

Exelon Generation.

U-604456

10 CFR 50.73
SRRS 5A.108

November 21, 2018

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2018-004-00

Enclosed is Licensee Event Report (LER) 2018-004-00: Undervoltage Condition Caused by an Offsite Ameren Transformer Bushing Failure Initiates an Automatic Start of an Emergency Diesel Generator. This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. Dale Shelton, Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Bradley T. Kapellas
Plant Manager
Clinton Power Station

KP/lam

Attachment: License Event Report 2018-004-00

cc:
Regional Administrator – Region III
NRC Senior Resident Inspector — Clinton Power Station
Office of Nuclear Facility Safety — Illinois Emergency Management Agency

IE22
NRR

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Clinton Power Station, Unit 1	05000461	1 OF 4

4. Title
Undervoltage Condition Caused by an Offsite Ameren Transformer Bushing Failure Initiates an Automatic Start of an Emergency Diesel Generator

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number 05000
09	26	2018	2018	- 004 -	00	11	21	2018	Facility Name	Docket Number 05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
099	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
Mr. Dale Shelton, Regulatory Assurance Manager	(217) 937-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES

14. Supplemental Report Expected	15. Expected Submission Date	Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No				

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On September 26, 2018, at 0946 CDT, the Clinton Power Station (CPS) experienced a trip of the Emergency Reserve Auxiliary Transformer (ERAT) Static VAR Compensator (SVC). A fault on the 138 kV feed to the station created a voltage transient which the ERAT SVC could not compensate for causing it to trip. Approximately 15 seconds following the trip of the ERAT SVC, the Division (DIV) 1 Safety Related 4160 kV Bus 1A1 locked out the Reserve Auxiliary Transformer (RAT) feed breaker and the ERAT feed breaker causing the DIV 1 Diesel Generator to start and load to the 1A1 bus due to a sustained degraded (second level) undervoltage condition as designed. Operations entered Technical Specification (TS) Limiting Condition for Operation (LCO) 3.8.1, "AC Sources-Operating" Required Actions A.1 and A.2 as a result of the voltage transient. The cause of the event was a failure of the high side bushing on offsite Clinton Route 54 to Substation Transformer No. 2 which caused the south feed to the Tabor Substation to cycle. Cycling of the south feed of the Tabor Substation caused the transient on the station 138 kV feed, which resulted in the subsequent station equipment responses. Ameren bypassed the affected transformer and restored the station 138 kV feed. The ERAT was restored to normal configuration on September 26, 2018, at 1354 CDT. Operations subsequently exited TS LCO 3.8.1, Actions A.1 and A.2. Ameren has completed installation of a 138 kV ring bus to improve the reliability of the 138 kV feed to CPS and is in the process of upgrading the feeds to the new substation. This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as any event or condition that resulted in manual or automatic actuation of emergency electrical power systems.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Clinton Power Station, Unit 1	05000461	2018	- 004	- 00

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric -- Boiling Water Reactor, 3473 Megawatts Thermal Rated Core Power
Energy Industry Identification System (EIIS) codes are identified in text as [XX].

EVENT IDENTIFICATION

Undervoltage Condition Caused by an Offsite Ameren Transformer Bushing Failure Initiates an Automatic Start of an Emergency Diesel Generator

A. Plant Operating Conditions Before the Event

Unit: 1	Event Date: September 26, 2018	Event Time: 0946
Mode: 1	Mode Name: Power Operation	Reactor Power: 099 percent

B. Description of Event

On September 26, 2018, at 0946 CDT, the Clinton Power Station (CPS) experienced a trip of the Emergency Reserve Auxiliary Transformer (ERAT) Static VAR Compensator (SVC)[COMP]. A fault on the 138 kV feed created a voltage transient ERAT SVC causing it to trip. Approximately 15 seconds following the trip of the ERAT SVC, the Division (DIV) 1 Safety Related 4160 kV Bus 1A1 locked out the Reserve Auxiliary Transformer (RAT) feed breaker and the ERAT feed breaker causing the DIV 1 Emergency Diesel Generator (EDG) to start and load to the 1A1 bus due to a sustained degraded (second level) undervoltage condition. Operations entered Technical Specifications (TS) Limiting Condition for Operation (LCO) 3.8.1, "AC Sources-Operating" Required Actions A.1 and A.2 as a result of the voltage transient.

A review of the safety related equipment actuation, and the response of the RAT, ERAT, ERAT SVC, and EDG, determined that plant equipment responded as designed for the event. Only the DIV 1 safety related 4160 kV Bus 1A1 was impacted since DIV 2 and 3 were being supplied by the RAT at the time of the event.

On September 26, 2018 at 1354 CDT the ERAT SVC was returned to Operable status following restoration of the offsite 138 kV line by Ameren and TS 3.8.1 Actions were exited.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Clinton Power Station, Unit 1	05000461	2018	- 004	- 00

NARRATIVE

C. Cause of the Event

The cause of the event was a failure of the high side bushing on offsite Clinton Route 54 Substation Transformer No. 2 which caused the south feed to the Tabor Substation to cycle. Cycling of the south feed to the Tabor Substation caused the transient on the station 138 kV feed, which resulted in the subsequent station equipment responses.

D. Safety Consequences

This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as any event or condition that resulted in manual or automatic actuation of emergency electrical power systems. The failure occurred at an offsite Ameren substation and did not directly involve CPS equipment.

The condition which caused a trip of the 138 kV supply line is bounded by the analysis in the Updated Safety Analysis Report section 15.2.6, "Loss of AC Power." The condition described in this report is less severe and involved no safety consequences since it only involved a loss of offsite power supply and not a trip of the unit. Electrical power sources are designed with sufficient redundancy to ensure the availability of necessary power to plant systems, structures, and components. All station equipment responded as designed to this event with no malfunctions.

E. Corrective Actions

The ERAT was restored to normal configuration at 1354 CDT on September 26, 2018. Ameren has completed installation of a 138 kV ring bus (Tabor Station) to improve the reliability of the 138 kV feed to CPS and is in the process of upgrading the feeds to the new substation.

NARRATIVE

F. Previous Similar Occurrences

There were no previous occurrences involving the loss of 138 kV line and ERAT trip due to failure of an offsite transformer. However, several similar occurrences involving loss of the 138 kV line have occurred. These events are summarized below.

Licensee Event Report 2017-001-00: Failure of the 138 kV Offsite Power Source Results in a Loss of Secondary Containment Vacuum

On February 24, 2017, at approximately 2239 hours CDT the ERAT and the associated SVC tripped due to a loss of the 138 kV line owned and maintained by Ameren Illinois. The 138 kV line was successfully re-energized at 0053 hours on February 25, 2017 and the ERAT off-site source was restored and declared OPERABLE at 0300 hours. The cause of this event was identified by Ameren as an air flow spoiler failure.

Licensee Event Report 2016-004-00: Trip of Emergency Reserve Auxiliary Transformer Static VAR Compensator Causes Positive Secondary Containment Pressure Following Lightning Strike on 138 kV Offsite Source

On March 30, 2016, at 1545 CDT, the ERAT SVC [COMP] tripped when the 138 kV line cycled open and closed due to a lightning strike. During the event, Division 1 Secondary Containment (SC) isolation dampers went shut due to the momentary loss of power and SC vacuum exceed 0 inches water gauge (WG). EOP-8 and TS LCO 3.6.4.1, Required Action A.1 were entered as a result of this abnormal condition. At 1547, the Standby Gas Treatment System (VG) Train A was started per CPS procedures. At 1550, SC vacuum had been restored to TS limits. Procedure EOP-8 and TS LCO 3.6.4.1 Required Action A.1 were exited. The ERAT was restored at 1926 hours and TS LCO 3.8.1 Required Actions A.1 and A.2 were exited.

The event on September 26, 2018 did result in a momentary reduction in SC vacuum. However, due to an approved change to the CPS TS, that event did not require declaring SC inoperable or entry into associated TS Actions.

G. Component Failure Data

The failed component that caused the fault on the 138 kV feed and subsequent ERAT trip is owned by Ameren. There were no CPS component failures associated with this event.